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Structural Privacy Vulnerabilities in Quantum Neural Networks

 (/pdf?id=T3eeSNIFUz)

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Abstract:

Quantum neural networks (QNNs) are increasingly considered for near-term learning tasks involving sensitive data, yet the influence of architectural design choices on their privacy risk remains poorly understood. Prior work has shown that QNNs can leak membership information, but it remains unclear where this leakage originates within the QNN pipeline. Is privacy risk primarily a consequence of the trainable ansatz, or can it be shaped upstream by the non-trainable encoder and its induced Hilbert-space geometry? We study this question through QuRiFT (Quantum Risk and Inference Fault-line Tracer), a controlled audit framework for structural privacy analysis in QML. Our central hypothesis is that the classical-to-quantum encoder shapes the Hilbert-space geometry on which the variational ansatz operates, inducing train-test asymmetries that later become exploitable by membership inference attacks. Using QuRiFT, we perform controlled interventions across QNN, HQNN, and QCNN models by varying feature-map family, data re-uploading repetitions, circuit width, variational depth, and variational-layer design while keeping training protocols fixed. The results reveal a clear structural hierarchy in our controlled regimes: privacy risk is driven more strongly by the data-encoding circuit than by trainable variational depth. In particular, z and zz feature maps induce larger generalization gaps and stronger membership signals than effSU2 . Feature-map repetitions and circuit width act as leakage amplifiers, with repetitions correlating strongly with larger train-test gaps ($r = 0.74$) and lower test accuracy ($r = -0.83$), producing a privacy-utility anti-tradeoff. Direct membership inference validates the structural signal: high-risk configurations identified by QuRiFT reach attack AUCs up to 0.870, compared with 0.530 for low-risk baselines. These findings show that the encoder should be treated as a first-order privacy parameter in QML design, with feature-map choice and data re-uploading audited as security-relevant architectural decisions rather than routine expressivity knobs.

Checklist Confirmation:  I confirm that I have included a paper checklist in the paper PDF.

Financial Support:  Najeeb Ullah (/profile?id=~Najeeb_Ullah1)

Responsible Reviewing:  We acknowledge the responsible reviewing obligations as authors.

Primary Area: General machine learning (e.g., core contributions in supervised and unsupervised methods)
Secondary Area: Deep learning advancements (e.g., architectures, optimizers, representation learning)
Contribution Type: Theory: The main contribution is via theoretical analyses and proofs.
Academic Integrity:  I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (<https://neurips.cc/Conferences/2026/MainTrackHandbook> (<https://neurips.cc/Conferences/2026/MainTrackHandbook>)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage:  Editing (e.g., grammar, spelling, word choice)
Declaration:  I confirm that the above information is accurate.
Reviewer Nomination:  Muhammad Naveed Aman (/profile?id=~Muhammad_Naveed_Aman1)
Submission Number: 24396

Discussion (?id=T3eeSNIFUz#discussion)

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Meta Review of Submission24396 by Area Chair zyrS

Meta Review by Area Chair zyrS  21 Jul 2026, 19:13 (modified: 24 Jul 2026, 02:16)
 Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs, Area Chair zyrS
 Revisions (/revisions?id=aF0m7l8e4X)

Metareview:

The paper studies structural sources of membership leakage in quantum NNs and introduces an approach to examine how encoder and circuit choices affect privacy risk. Reviewers found the encoder-centric perspective interesting and appreciated the controlled empirical analysis. A major limitation, however, is that all experiments are conducted under noiseless simulation, whereas current quantum hardware is inherently noisy and noise may significantly alter, or even mitigate, the observed privacy leakage. It is therefore unclear whether the reported privacy hierarchy would persist under realistic noise conditions.

Reviewers also raised fundamental questions about the experiments and conclusions, for example whether encoder choices independently cause membership leakage or merely induce ordinary overfitting, whether the generalization gap is an adequate proxy for membership risk; whether the selected configurations and limited attack suite are representative; and whether the reported trends are statistically robust and generalize beyond the simple datasets and confounded architecture comparisons considered.

Also, the paper would benefit from a more thorough discussion of privacy in quantum ML and a clearer presentation of recent related work, which would better contextualize the contribution and improve the exposition.

All these concerns limit the contributions and appear to require major revision.

Add: **Official Comment** **Author AC Confidential Comment**



Official Comment

Official Comment by 31 Jul 2026, 04:43 (modified: 31 Jul 2026, 04:45)

Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs

Revisions (/revisions?id=y5bahpuF5o)

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Official Review of Submission24396 by Reviewer Epmi

Official Review by Reviewer Epmi 26 Jun 2026, 19:58 (modified: 29 Jul 2026, 02:54)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Epmi

Revisions (/revisions?id=RgUIjAtDd8)

Summary:

This paper studies structural sources of membership leakage in quantum neural networks. The main hypothesis is that the non-trainable classical-to-quantum encoder shapes the Hilbert-space representation before optimization, and that this encoder-induced structure can create train-test asymmetry that later becomes exploitable by membership inference methods. To study this, the paper introduces QuRiFT, a controlled profiling framework that sweeps feature-map family, feature-map repetitions, circuit width, variational depth, entanglement topology, and gate choices across QNN, HQNN, and QCNN models. The paper reports that Z/ZZ feature maps and larger feature-map repetitions tend to produce larger generalization gaps than efficient-SU(2)-style encoders, and that selected high-risk configurations can lead to higher MIA AUC.

The topic is interesting, and the paper raises a useful question: whether QML privacy risk should be treated as a structural design issue rather than only as a consequence of training dynamics. However, I am not fully convinced that the current evaluation isolates the proposed structural source of leakage from generic training-induced leakage due to overfitting.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

1. Interesting problem framing. The paper studies membership leakage from a QML architecture-design perspective. Instead of only asking whether QNNs leak membership, it asks which circuit components may induce the leakage. This is a meaningful and potentially useful angle.
2. Useful controlled profiling framework. QuRIFT separates feature-map family, feature-map repetitions, circuit width, variational depth, and lower-level variational-layer choices under a fixed training protocol. The distinction between repeated data encoding and variational depth is especially useful, because repetitions inject input-dependent structure through the fixed encoder, whereas variational depth mainly changes trainable capacity.

Weaknesses

1. The evaluation does not sufficiently isolate the proposed structural source of leakage from generic training-induced leakage. The central claim is that encoder-side structural choices, especially feature-map family and data-reuploading repetitions, are important sources of membership leakage. However, the current evidence mostly shows that some structural configurations have larger train-test gaps, and that selected high-gap configurations can have higher MIA AUC. This is not enough to establish that the MIA signal is caused by the proposed structural factors rather than by ordinary overfitting after training.

This distinction matters because membership inference methods exploit differences in output behavior between training and non-training samples. Since the attack features used in this paper include loss, confidence, margin, entropy, and correctness, a higher MIA AUC may simply reflect a larger train-test gap or poorer calibration. In other words, the current results show that some trained models leak more, but they do not cleanly show that the encoder/repetition choices independently cause the leakage.

A stronger evaluation would include pairwise comparisons where all other factors are fixed or matched, and only the proposed structural variable changes. For example, compare two models with the same architecture, same width, same variational depth, same training protocol, similar train accuracy, similar test accuracy, and similar generalization gap, but different feature-map family or repetition count. If the MIA AUC still differs, then the claim that these structural choices independently affect membership leakage would be much stronger. Conversely, if the difference disappears after matching the gap, then the contribution should be reframed as identifying structural factors that induce overfitting, not as directly validating structure-induced membership leakage.

2. The paper relies heavily on the generalization gap as a privacy-risk proxy, but generalization gap is not equivalent to membership leakage. The train-test accuracy gap is a useful first-order indicator of overfitting, but it is a dataset-level aggregate metric. MIA is a sample-level inference problem and depends on the separability of member and non-member output distributions, confidence tails, calibration, class-specific behavior, and the specific attack features. Therefore, a large gap does not necessarily imply strong MIA, and a small gap does not rule out membership leakage on rare samples, hard examples, or specific classes.

The paper itself reports exceptions where the gap is not a deterministic predictor of MIA success. This weakens the interpretation of the selected stress/hard regimes. The authors should analyze the relationship between MIA AUC and generalization gap more directly. For example, they could show MIA AUC versus generalization gap across a broader

set of configurations, mark feature-map family and repetition count, and explicitly check whether configurations with similar gaps can have different MIA AUCs, or whether some low-gap configurations still show non-trivial MIA. This would clarify whether feature-map choice and repetitions affect membership leakage beyond simply increasing the train-test gap.

3. The MIA evaluation is limited to one access setting and one attack-feature family. The paper evaluates what the authors describe as a black-box MIA, but this setting still assumes access to the full prediction vector. The attack uses features derived from confidence, entropy, margin, loss, and correctness when the label is known. This is a valid score/probability-vector access setting, but it does not cover the broader range of MIA methods that may emphasize different leakage signals.

In particular, the paper does not separately report simple threshold methods based on loss, confidence, entropy, or margin. This makes it hard to tell which output signal is actually responsible for the reported MIA success. It also does not evaluate calibrated methods using reference or shadow models, which can behave differently from raw score-based attacks, or label-only methods, which test a weaker API setting where probabilities are unavailable.

This matters because different MIA methods measure different properties of the model. Loss-based methods are closely tied to train/test loss separation; confidence and entropy methods are affected by calibration and posterior sharpness; reference-model methods compare the target output against in/out distributions; and label-only methods depend more on decision-boundary robustness. Therefore, the current results do not show whether the reported structural privacy trends are robust across attack settings, or whether they are specific to the chosen full prediction-vector feature set.

I recommend adding separate threshold results for loss, confidence, entropy, and margin, and adding at least one calibrated reference/shadow-model method or one label-only method if feasible. Reporting both AUC and low-FPR TPR for these methods would clarify which membership signals are actually affected by feature-map choice and repetitions.

4. The selected MIA regimes and main trends need stronger statistical support. The baseline/stress/hard regimes are useful for illustration, but they are selected after observing structural overfitting patterns. This makes it difficult to assess whether the observed attack AUC is representative across the full design space. A more convincing approach would run the same MIA suite over all configurations, followed by a regression analysis of attack AUC. The regression should include feature-map family, repetitions, width, depth, train accuracy, test accuracy, generalization gap, gate count, and number of trainable parameters. In addition, the paper should report uncertainty for the main plots, attack AUCs, correlations, and factor-attribution results. Error bars, confidence intervals, or multi-seed/multi-split results are important here because the reported trends may be sensitive to train/test splits, initialization, attack-model training, and the selected configurations.
5. The central Hilbert-space geometry mechanism remains indirect. The paper argues that the encoder induces a Hilbert-space geometry that leads to train-test asymmetry and membership leakage. However, the experiments mainly measure downstream quantities such as generalization gap, attack AUC, and factor correlations. These are useful indicators but do not directly show how the induced kernel geometry changes. The paper would be stronger with direct measurements of train-train, train-test, and test-test Hilbert-Schmidt similarity distributions, kernel alignment with labels, nearest-neighbor concentration, kernel spectrum/effective rank, or margin distributions under different feature maps and repetition counts.

6. The architecture-level and simulation settings limit the scope of the conclusions. The QNN, HQNN, and QCNN settings differ not only in quantum architecture but also in preprocessing and classical heads. Dense QNN uses non-trainable compression, HQNN uses a trainable CNN bottleneck, and QCNN uses a quantum-filter front-end plus an MLP head. Therefore, the observed dampening effect may come from the classical preprocessor, the head capacity, or the latent representation, not only from quantum architectural inductive bias. The paper should report parameter counts and add matched-parameter or frozen-preprocessor ablations.

I also agree that noiseless simulation is useful for isolating encoder effects. However, repetitions and ZZ-style encoders change circuit depth and gate count, and hardware noise or finite-shot effects could change both utility and privacy behavior. The paper should either limit its conclusions to noiseless controlled regimes or include a small finite-shot/noise-model sanity check.

Quality: 3: good

Clarity: 2: not good

Significance: 3: good

Originality: 3: good

Questions:

1. Can the authors provide a clean pairwise ablation where all settings are fixed or matched, and only the feature-map family or repetition count changes before measuring MIA AUC?
2. Can the authors analyze the relationship between generalization gap and MIA more directly? In particular, are there configurations with similar train/test behavior or similar generalization gap but noticeably different MIA AUC, or configurations with small generalization gap but non-trivial MIA risk? Such an analysis would clarify whether the proposed structural factors affect membership leakage beyond simply increasing the train-test gap.
3. Can the authors clarify how representative the selected baseline/stress/hard regimes are within the full sweep? Additional MIA results on a broader subset of configurations, together with uncertainty estimates for the main attack AUCs, would help assess whether the reported trends are stable.
4. Which specific attack feature drives the reported MIA success? Is it loss, confidence, entropy, margin, correctness, or their combination? Please report separate threshold methods for these signals and compare with calibrated reference-model, shadow-model, or label-only methods if feasible.
5. Can the authors directly measure the induced Hilbert-space geometry, for example through kernel similarity distributions, kernel alignment, nearest-neighbor concentration, or kernel spectrum, rather than relying mainly on the generalization gap?

Limitations:

The authors acknowledge several limitations, including the controlled noiseless simulation setting and the need for broader attacks and direct geometry measurements. This is useful, but the current limitations section does not fully address the main evaluation concern: the experiments do not yet isolate whether the proposed structural factors independently cause membership leakage, as opposed to merely inducing ordinary overfitting that is then exploited by an output-based MIA.

The limitations section should explicitly discuss this causal-attribution issue. It should also clarify that the current black-box setting assumes access to score/probability vectors, and therefore should not be interpreted as covering label-only or query-limited black-box settings. The current conclusions may also not transfer directly to calibrated/reference-

model attacks or noisy quantum hardware.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

None

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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by Authors (👁 Biplab Sikdar (/profile?id=~Biplab_Sikdar1), Najeed Ullah (/profile?id=~Najeed_Ullah1), Muhammad Naveed Aman (/profile?id=~Muhammad_Naveed_Aman1))

📅 01 Aug 2026, 01:27

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Epmi

Comment:

We deeply appreciate your comprehensive feedback, which greatly improved our work.

Our submission reported a broad exploratory sweep of 349920 configs across datasets, architectures, encoders, repetitions, depths, and lower-level circuit choices. For the rebuttal, we added a focused MNIST-QNN factorial (3 feature maps $\times$ 2 repetitions $\times$ 2 depths $\times$ 3 target seeds = 36 targets) and applied the same attack suite to every target. A detailed response can be found here (<https://github.com/CartwheelX/QuRiFT/blob/main/docs/Reviewer-epmi.md> (<https://github.com/CartwheelX/QuRiFT/blob/main/docs/Reviewer-epmi.md>)).

Which output signal drives MIA?

We evaluated every requested scalar separately. Mean AUC $\pm$ sample SD across 36 targets is: loss 0.596 ± 0.052 ; correctness 0.582 ± 0.049 ; entropy 0.521 ± 0.023 ; margin 0.521 ± 0.026 ; and confidence/maximum probability 0.520 ± 0.025 . Confidence and maximum probability are the same exported scalar, not independent attacks. The learned prediction-vector-plus-statistics attacker obtains 0.574 ± 0.065 . Thus, loss is the clearest raw signal, and combining features does not improve its mean AUC. Correctness has zero TPR at 5% and 10% FPR because its score is binary.

Fixed-variance online LiRA is numerically strongest overall (AUC 0.609 ± 0.063 ; TPR@5% FPR 0.128 ± 0.056). It uses 16 same-architecture references per structural configuration; each candidate is IN for eight. The label-only obtains 0.582 ± 0.052 AUC and 0.077 ± 0.027 TPR@5% FPR.

Matched-pair structural effects across the calibrated and label-only attacks.

Attack	Factor	Contrast	AUC difference $\pm$ SD	95% CI	Paired units
Online LiRA, fixed variance	Repetitions	5 – 1	+0.044 $\pm$ 0.045	[0.013, 0.072]	18
Online LiRA, fixed variance	Depth	6 – 2	+0.077 $\pm$ 0.057	[0.038, 0.118]	18
Online LiRA, fixed variance	Feature map	Z – EffSU2	+0.057 $\pm$ 0.030	[0.038, 0.076]	12
Online LiRA, fixed variance	Feature map	ZZ – EffSU2	+0.070 $\pm$ 0.059	[0.021, 0.117]	12
Online LiRA, fixed variance	Feature map	ZZ – Z	+0.014 $\pm$ 0.044	[-0.026, 0.048]	12
Label-only	Repetitions	5 – 1	+0.068 $\pm$ 0.026	[0.049, 0.085]	18
Label-only	Depth	6 – 2	+0.047 $\pm$ 0.033	[0.021, 0.068]	18
Label-only	Feature map	Z – EffSU2	+0.038 $\pm$ 0.035	[0.004, 0.061]	12
Label-only	Feature map	ZZ – EffSU2	+0.054 $\pm$ 0.039	[0.022, 0.087]	12
Label-only	Feature map	ZZ – Z	+0.017 $\pm$ 0.026	[-0.004, 0.035]	12

These paired contrasts connect attack breadth to the structural claim. In each contrast, only the tested factor changes; all other settings are matched. Repetition raises online LiRA AUC in five of six fixed-depth comparisons and has a positive pooled effect. Z and ZZ exceed EffSU2 under both LiRA and label-only access, whereas ZZ minus Z remains unresolved. Depth also contributes, so we do not attribute leakage solely to the encoder. The supported conclusion is that encoder family and repetition change the representation presented to training and condition the downstream asymmetry exploited by different MIAs.

Direct Hilbert-space geometry.

We compute the fidelity kernel $K(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$ immediately after the fixed encoder and before the trainable variational circuit. Paired reps=5 minus reps=1 effects are:

Geometry measurement	Definition	Reps 5 – reps 1 $\pm$ SD	95% CI
Class-similarity gap	mean within-class minus between-class fidelity	-0.124 $\pm$ 0.061	[-0.158, -0.079]
Kernel-label alignment	centered fidelity-kernel/label-kernel alignment	-0.208 $\pm$ 0.132	[-0.285, -0.109]
Effective rank	spectral effective rank of the fidelity kernel	+49.747 $\pm$ 37.448	[15.791, 83.870]

Geometry measurement	Definition	Reps 5 – reps 1 ±	
		SD	95% CI
Train/test MMD ²	kernel discrepancy between train and test encoded states	−0.003 ± 0.008	[−0.010, 0.003]

For paired reps=5 minus reps=1 contrasts, the within-class minus between-class similarity gap changed by -0.124 ± 0.061 , with a 95% hierarchical-bootstrap CI of $[-0.158, -0.079]$; kernel-label alignment changed by -0.208 ± 0.132 , CI $[-0.285, -0.109]$; and effective rank increased by 49.747 ± 37.448 , CI $[15.791, 83.870]$. These observations support the first link in our proposed pathway: increasing repetition changes the post-encoder representation toward a higher-rank but less class-aligned geometry.

The train-test MMD² contrast was -0.003 ± 0.008 , CI $[-0.010, 0.003]$, and is interpreted as approximately zero. Thus, the encoder changes task geometry but does not directly separate member and non-member populations before training.

Overall, across the 36 targets, accuracy gap strongly tracks loss-AUC (Spearman $\rho=0.931$, 95% CI $[0.710, 0.974]$). After conditioning descriptively on gap, the residual structural coefficient intervals cross zero. We therefore refine the claim to an empirically supported, overfitting-mediated pathway: encoder design → post-encoder geometry → downstream generalization asymmetry → membership signal.

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Official Review of Submission24396 by Reviewer 1myw

Official Review by Reviewer 1myw 📅 26 Jun 2026, 13:18 (modified: 29 Jul 2026, 02:54)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 1myw

📄 Revisions (/revisions?id=z7omMThoJ2)

Summary:

The paper mainly argues that the non-trainable data encoders in QNNs expose a vulnerable attack surface for privacy leakage, such as membership inference attacks (MIAs). Starting from the abstract itself, the authors make it explicitly clear that certain encoding methods, like Z and ZZ feature maps, along with repeated data re-uploading, increased the vulnerability to MIAs. This vulnerability is claimed to be induced by train-test asymmetries, which are methodically tested via the termed "QuRiFT" framework. So far, the paper presents the experimental validations showing the leakages.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths:

1. The core idea of studying privacy leakages in the very first encoding stage of a QNN, rather than the VQC training part (some related works have done it), is a novel direction.
2. The experimental setup is good with a lot of ablations around encoders, width, entangling ways and gate types. The experiments are controlled and the overall paper is easy to follow.
3. The authors provide empirical evidence of a privacy-utility anti-tradeoff, which was particularly interesting. Also, from the negative-control analysis, it is good to see that the lower-level layers have a lesser effect than the encoding layers.
4. The validation experiments with black-box membership inference attacks with prediction-vector features are coherent with the structural vulnerability claims. For a preliminary qualitative validation, the experiments are largely complete.

Weaknesses:

1. Major: Noiseless Simulation The authors acknowledge this as a limitation of their work and tend to justify the choice of noiseless simulation as a method to get isolated results in their claim. However, opting noise out in this scenario raises concerns, particularly when data leakage is involved. Although I agree that an ideal simulation to understand the fundamental structural vulnerability is essential, without realistic experiments including noise, the contribution is weak and partial, more like an ablation with initial findings. Interestingly, quantum noise channels have been proposed multiple times to provide different privacy mechanisms and adversarial robustness in QNNs. The severity and even dynamics of vulnerability might be different under these scenarios.
2. A limitation as well as a weakness is the statement that the structural privacy hierarchy identified may not generalize to all possible QML architectures, datasets, or hardware backends.
3. The MIA evaluation seems too narrow for a NeurIPS grade paper. The black box threat model and output-based design are valid for a specific MIA case, but the attack suite still seems limited. Baselines such as threshold attacks, shadow model-based, or maybe LiRA style would strengthen the paper.
4. Datasets considered are too simple.

Quality: 3: good

Clarity: 3: good

Significance: 2: not good

Originality: 3: good

Questions:

Since all experiments are done in noiseless simulation, can the authors show whether the same privacy hierarchy still holds under realistic quantum noise or finite-shot settings?

The paper states that the structural privacy hierarchy may not generalize to all QML architectures, datasets, or hardware backends. Can the authors clarify the exact scope of this hierarchy and which findings they expect to generalize?

Since the datasets are relatively simple, can the authors clarify how far the conclusions can be extended to the sensitive domains used in the motivation, such as healthcare, finance, and public-sector data?

Also, please consider my concerns from the weakness section.

Limitations:

yes.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:

An overview figure would help.

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

Add:

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Author AC Confidential Comment



Geometric Leakage Mechanism Is Unverified and Potentially Noise-Dependent

Official Comment by Reviewer 1myw 📅 31 Jul 2026, 04:34

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 1myw

Comment:

As is, the authors have acknowledged the noiseless setting as a limitation and justify the choice as isolated evaluation for the claimed structural effect. It is fine to study fundamental vulnerabilities in isolated setting. However, in this case, excluding noise is not only a matter of limited validation scope. Especially NISQ noise acts directly on the encoded quantum state and the paper identifies this as the source of the privacy hierarchy. Consequently, such uncertainty may reorder the observed hierarchy rather than shift all configurations uniformly. Even a cheap finite-shot experimental support would provide a real robustness check.

More importantly, limited scope might be okay for the declared contribution type, but the geometric mechanism itself is not directly measured. K_ϕ and the related ones in Appendix D are defined but not computed or rigorously analyzed. As presented, the main evidence for the mechanism is the accuracy gap, followed by membership-inference evaluation on a selected configurations. This is not usually enough to distinguish the geometric explanation from the simpler ones that certain circuit configurations merely overfit more than others.

The relevance of this concern is taken further as quantum noise channels have previously been considered as potential mechanisms for privacy protection and adversarial robustness in QNNs. The severity, ordering, and dynamics of the reported vulnerabilities may therefore differ under noisy execution.

As the paper currently stands (without rebuttal and overall discussions), I intend to lower the Quality score from 3 to 2 in my final recommendation. I will look forward to the rebuttal and clarification of the raised concerns before changing any other scores.

Add: Official Comment Author AC Confidential Comment



Official Comment by Authors

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Official Comment

by Authors (👁️ Biplab Sikdar (/profile?id=~Biplab_Sikdar1), Najeeb Ullah (/profile?id=~Najeeb_Ullah1), Muhammad Naveed Aman (/profile?id=~Muhammad_Naveed_Aman1))

📅 31 Jul 2026, 22:17

👁️ Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 1myw

Comment:

Thank you for the follow-up.

We agree that the original submission did not directly measure the geometry discussed in Appendix D. In response, we began multi-seed geometry and noisy finite-shot experiments. These required trained checkpoints and repeated circuit simulations across shot counts, seeds, and noise conditions, so they took longer than a standard ablation. The geometry and noise analyses are now complete. for detailed discussion and tables (see <https://github.com/CartwheelX/QuRiFT/blob/main/docs/Reviewer-1myw.md> (<https://github.com/CartwheelX/QuRiFT/blob/main/docs/Reviewer-1myw.md>)).

Backend-derived noise and finite shots. For a quick sanity check, we evaluated five representative configurations varying feature-map family, encoder repetitions, and variational depth. Their exact-inference loss-MIA AUCs ranged from 0.521 to 0.700. Each was trained with three target-model seeds, giving 15 checkpoints. We evaluated the same member and non-member records using exact inference, ideal finite-shot Aer, and Aer with a noise model derived from the `ibm_kingston` calibration. Shot-based evaluations used 128, 512, and 1,024 shots with ten simulator seeds.

Finite-shot uncertainty and backend-derived noise reduced membership leakage, but did not remove the association between structural configuration and mean attack AUC in this subset. Configurations with weaker leakage under exact inference continued to show weaker mean leakage, while those with stronger leakage continued to show stronger mean leakage. The AUC difference between ZZ with reps=5, depth=6 and EfficientSU2 with reps=1, depth=2 decreased from 0.179 ± 0.030 under exact inference to 0.096 ± 0.012 at 128 noisy shots, then increased to 0.124 ± 0.026 and 0.132 ± 0.036 at 512 and 1,024 shots.

Individual simulator runs sometimes exchanged the relative positions of configurations with similar AUCs, particularly at 128 shots. Mean Spearman correlation with the exact-inference results was 0.820 ± 0.155 , 0.920 ± 0.092 , and 0.960 ± 0.052 at 128, 512, and 1,024 noisy shots. Thus, noise mitigates leakage and adds local variability, but does not remove the association between structural configuration and mean attack AUC in this evaluation. This supports our main claim that encoder design and repetition condition downstream membership risk.

Direct post-encoder geometry. We further compute the pure-state fidelity kernel immediately after the fixed encoder and before the variational circuit. For paired reps=5 minus reps=1 contrasts, the within-class minus between-class similarity gap changed by -0.124 ± 0.061 , with a 95% hierarchical-bootstrap CI of $[-0.158, -0.079]$; kernel-label alignment changed by -0.208 ± 0.132 , CI $[-0.285, -0.109]$; and effective rank increased by 49.747 ± 37.448 , CI $[15.791, 83.870]$. These observations support the first link in our proposed pathway: increasing repetition changes the post-encoder representation toward a higher-rank but less class-aligned geometry.

The train-test MMD² contrast was -0.003 ± 0.008 , CI $[-0.010, 0.003]$, and is interpreted as approximately zero. Thus, the encoder changes task geometry but does not directly separate member and non-member populations before training.

We agree with the reviewer's causal distinction. Our claim is not that geometry bypasses overfitting or directly reveals membership. The evidence supports associations along the pathway **encoder design → post-encoder geometry → downstream generalization asymmetry → membership signal**. The kernel analysis supports the encoder-to-geometry association. The factorial results support repetition-to-gap and repetition-to-loss-AUC associations. The generalization gap strongly tracks loss-MIA AUC, with hierarchical-bootstrap Spearman $\rho=0.931$, CI $[0.710, 0.974]$. Together, these results support an encoder-induced, overfitting-mediated account of membership risk. We will add these results and tables to the revision and replace causal wording with "empirically supported geometric pathway."

Attack breadth and scope. We have completed threshold attacks using loss, confidence, entropy, margin, maximum probability, and correctness. We are also running LiRA reference-model and label-only baselines. These complement the original learned prediction-vector attacker and test whether the signal persists across adversaries with different information access. We will post them shortly here in the comments.

Lastly, we thank the reviewer for prompting these analyses. They materially improved both the empirical validation and the precision of the paper's central claim.

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by Authors ( Biplab Sikdar (/profile?id=~Biplab_Sikdar1), Najeeb Ullah (/profile?id=~Najeeb_Ullah1), Muhammad Naveed Aman (/profile?id=~Muhammad_Naveed_Aman1))

 01 Aug 2026, 02:01 (modified: 01 Aug 2026, 02:02)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 1myw

 Revisions (/revisions?id=N8wVZjA6Fw)

Comment:

This is a follow-up to the already-posted geometry/noise response.

Following up on the attack-breadth paragraph in our previous response, the reference-model LiRA and class-label-only experiments are now complete. Both were evaluated on all 36 targets in our focused confirmatory MNIST-QNN factorial (12 structural configurations $\times$ three independently initialized target models). This is the multi-seed follow-up subset, not a claim that we reran every configuration in the submission's broader architecture/dataset sweep.

For LiRA, we trained 16 same-architecture reference QNNs per structural configuration (192 references total). Every one of the 400 candidate records was included in exactly eight references, and every reference was trained on 200 candidates. The primary fixed-variance online LiRA result is 0.609 ± 0.063 AUC, with TPR 0.128 ± 0.056 at 5% FPR and 0.192 ± 0.069 at 10% FPR. The per-record-variance online result is 0.594 ± 0.057 AUC, showing that the result is not specific to one variance model. The offline result is much weaker, which we report rather than selecting only the strongest LiRA variant.

The label-only attack consumes only returned class labels and estimates input-space boundary distance by changed-label searches toward held-out validation anchors. It obtains 0.582 ± 0.052 AUC, with TPR 0.077 ± 0.027 at 5% FPR and 0.139 ± 0.030 at 10% FPR.

Attack	Information access	AUC	TPR@5% FPR	TPR@10% FPR
Loss threshold	true-label probability + known candidate label	0.596 ± 0.052	0.061 ± 0.024	0.135 ± 0.039
Learned prediction-vector	full prediction vector + statistics	0.574 ± 0.065	0.099 ± 0.044	0.157 ± 0.061
Online LiRA, fixed variance	true-label probability + calibrated reference QNNs	0.609 ± 0.063	0.128 ± 0.056	0.192 ± 0.069
Online LiRA, per-record variance	true-label probability + calibrated reference QNNs	0.594 ± 0.057	0.086 ± 0.034	0.155 ± 0.063
Offline LiRA, fixed variance	true-label probability + calibrated reference QNNs	0.517 ± 0.029	0.066 ± 0.027	0.128 ± 0.035

Attack	Information access	AUC	TPR@5% FPR	TPR@10% FPR
Label-only	predicted class labels only + held-out anchors	0.582 ± 0.052	0.077 ± 0.027	0.139 ± 0.030

Entries are mean ± sample SD across 36 target models. The learned attack is first averaged across three attacker seeds per target. LiRA uses one 16-reference bank per structural configuration.

The paired analysis supports the encoder-induced pathway across access settings. Repetition increases label-only AUC by 0.068 ± 0.026 (95% CI [0.049, 0.085]) and pooled LiRA AUC by 0.044 ± 0.045 ([0.013, 0.072]). In fixed-depth comparisons, LiRA AUC rises with repetition in five of six configurations. Z and ZZ remain higher than EffSU2 for both attacks, while the ZZ–Z intervals include zero. These calibrated attacks therefore support the claim that encoder family and repetition condition the downstream membership signal.

Attack	Factor	Contrast	AUC difference ± SD	95% CI	Paired units
Online LiRA, fixed variance	Repetitions	5 – 1	+0.044 ± 0.045	[0.013, 0.072]	18
Online LiRA, fixed variance	Depth	6 – 2	+0.077 ± 0.057	[0.038, 0.118]	18
Online LiRA, fixed variance	Feature map	Z – EffSU2	+0.057 ± 0.030	[0.038, 0.076]	12
Online LiRA, fixed variance	Feature map	ZZ – EffSU2	+0.070 ± 0.059	[0.021, 0.117]	12
Online LiRA, fixed variance	Feature map	ZZ – Z	+0.014 ± 0.044	[-0.026, 0.048]	12
Label-only	Repetitions	5 – 1	+0.068 ± 0.026	[0.049, 0.085]	18
Label-only	Depth	6 – 2	+0.047 ± 0.033	[0.021, 0.068]	18
Label-only	Feature map	Z – EffSU2	+0.038 ± 0.035	[0.004, 0.061]	12
Label-only	Feature map	ZZ – EffSU2	+0.054 ± 0.039	[0.022, 0.087]	12

Attack	Factor	Contrast	AUC difference $\pm$		Paired units
			SD	95% CI	
Label-only	Feature map	ZZ – Z	+0.017 $\pm$ 0.026	[-0.004, 0.035]	12
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Official Review of Submission24396 by Reviewer nVBH

Official Review by Reviewer nVBH 📅 25 Jun 2026, 23:57 (modified: 29 Jul 2026, 02:54)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer nVBH

📄 Revisions (/revisions?id=RVsrSApndS)

Summary:

This paper studies structural privacy vulnerabilities in quantum neural networks. The central hypothesis is that the fixed classical-to-quantum encoder can shape the Hilbert-space geometry before optimization, inducing train-test asymmetries that later become exploitable by membership inference attacks. To investigate this, the paper introduces QuRiFT, a controlled auditing framework that varies feature-map family, feature-map repetitions, circuit width, variational depth, entanglement topology, and gate choices across QNN, HQNN, and QCNN models. The experiments are conducted on synthetic binary datasets and a compressed four-class MNIST subset. The paper reports that Z and ZZ feature maps, feature-map repetitions, and circuit width are more strongly associated with generalization gaps and membership-inference leakage than variational depth or lower-level variational-layer choices. The main conclusion is that encoder design and repeated data encoding should be treated as privacy-relevant architectural decisions in QML.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

1. Important and timely problem. The paper addresses privacy leakage in near-term quantum machine learning models, with a focus on membership inference. This is a relevant problem, especially because QML is often motivated by applications involving sensitive data.
2. Interesting encoder-centric framing. A useful aspect of the paper is that it does not treat membership leakage only as a consequence of optimization or trainable model capacity. Instead, it studies whether the fixed classical-to-quantum encoder itself can induce privacy-relevant train-test asymmetries. This is a meaningful perspective for QML model design.
3. Coherent controlled-audit framework. The proposed QuRiFT framework provides a systematic way to vary feature-map family, feature-map repetitions, circuit width, variational depth, entanglement topology, and gate choices under a controlled training setup. This makes the paper more than a single-model membership-inference study.

4. Suggestive empirical findings. The reported trends are potentially useful: Z and ZZ feature maps appear to produce larger generalization gaps than effSU2; feature-map repetitions correlate more strongly with overfitting and attack metrics than variational depth; and repeated encoding may reduce test accuracy while increasing leakage. These observations could be valuable for QML practitioners if made statistically robust.
5. Direct MIA validation. The paper does not rely only on generalization gaps. It also runs output-based black-box membership inference attacks, which is important for validating whether the proposed structural proxy corresponds to actual leakage.
- Weaknesses
6. The paper lacks multi-seed results, confidence intervals, error bars, and significance tests, making the main attack and trend claims difficult to assess for robustness.
7. Regime selection appears post hoc and overly tied to generalization gaps, creating circular validation and weakening claims that structural overfitting reliably predicts membership leakage.
8. Claims about feature maps, encoder geometry, and architectures as leakage amplifiers are correlational, with confounders such as capacity, preprocessing, convergence, and accuracy insufficiently controlled.
9. The study lacks classical and QML baselines, transparent factor-attribution methodology, and broader datasets, limiting attribution and generalization to sensitive-domain settings.

Quality: 2: not good

Clarity: 3: good

Significance: 2: not good

Originality: 3: good

Questions:

1. Can the authors provide multi-seed results with confidence intervals or error bars for the main figures and attack metrics? Are the trends for Z/ZZ versus effSU2, repetitions versus depth, and stress versus hard regimes stable across random initialization, data splits, and MIA attacker training?

2. Can the authors report membership-inference results across all swept configurations, rather than only selected baseline/stress/hard regimes? This would help determine whether the generalization gap is a reliable structural proxy or whether the current validation is affected by post-hoc selection.

3. Can the authors clarify how the factor-attribution percentages are computed? Please specify whether the method is based on ANOVA, Sobol-style variance decomposition, regression, random-forest importance, or another method, and provide uncertainty estimates or sensitivity checks.

4. Can the architecture comparison be made more controlled? For example, can the authors compare QNN, HQNN, and QCNN under matched parameter counts, matched preprocessing capacity, or with classical-only versions of the CNN/MLP components?

5. Can the authors add stronger baselines or controls, such as classical MLP/CNN models, classical kernel methods, quantum kernel classifiers, regularized QNNs, early stopping, differential privacy, or calibration-based defenses?

Limitations:

The authors discuss some limitations, such as noiseless simulation and the need for future work on broader attacks and direct geometry measurements. However, the limitations section does not adequately address several central limitations of the current study: insufficient statistical uncertainty, post-hoc regime selection, lack of classical and stronger QML baselines, confounded architecture comparisons, and narrow datasets relative to the sensitive-domain motivation. I recommend expanding the limitations section to explicitly acknowledge these issues and to weaken claims about sensitive-domain QML deployment unless more realistic datasets and baselines are added.

Rating: 3: Borderline reject: Technically solid paper where reasons to reject, e.g., limited evaluation, outweigh reasons to accept, e.g., good evaluation. Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Ethical Concerns: NO or VERY MINOR ethics concerns only

Paper Formatting Concerns:
N/A

Code Of Conduct Acknowledgement: Yes

Responsible Reviewing Acknowledgement: Yes

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Official Comment by Authors

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Official Comment by Authors 01 Aug 2026, 22:53
 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer nVBH

Comment:

We thank the reviewer for the careful reading and constructive questions.

Evaluating the full original sweep space of 349,920 configurations was not feasible due to computational constraints of the rebuttal. Instead, we added a focused complete MNIST-QNN factorial (3 feature maps × 2 repetitions × 2 depths × 3 target seeds = 36 targets) and applied the same attacks to every target without post-hoc selection. Full protocols, per-configuration results, and attribution equations are detailed here (<https://github.com/CartwheelX/QuRiFT/blob/main/docs/Reviewer-nVBH.md> (<https://github.com/CartwheelX/QuRiFT/blob/main/docs/Reviewer-nVBH.md>)).

1. Multi-seed robustness.

Values below are paired AUC differences ± SD with 95% hierarchical-bootstrap CIs. Each configuration has three target seeds; the learned attacker has three training seeds, and LiRA uses 16 references per configuration. The data split is fixed, so these isolate structural effects.

Attack	Contrast	AUC diff ± SD	95% CI	Paired Units
Loss	Reps 5 – 1	+0.069 ± 0.029	[0.049, 0.088]	18
Loss	Depth 6 – 2	+0.042 ± 0.026	[0.024, 0.060]	18
Loss	Z – EffSU2	+0.057 ± 0.032	[0.027, 0.080]	12
Loss	ZZ – EffSU2	+0.052 ± 0.027	[0.030, 0.074]	12
Online LiRA	Reps 5 – 1	+0.044 ± 0.045	[0.013, 0.072]	18
Online LiRA	Depth 6 – 2	+0.077 ± 0.057	[0.038, 0.118]	18

Attack	Contrast	AUC diff $\pm$ SD	95% CI	Paired Units
Online LiRA	Z — EffSU2	+0.057 $\pm$ 0.030	[0.038, 0.076]	12
Online LiRA	ZZ — EffSU2	+0.070 $\pm$ 0.059	[0.021, 0.117]	12
Online LiRA	ZZ — Z	+0.014 $\pm$ 0.044	[−0.026, 0.048]	12
Label-only	Reps 5 — 1	+0.068 $\pm$ 0.026	[0.049, 0.085]	18

These results support the claim that encoder family and repetition condition leakage across target initializations and access models.

2. Post-hoc selection and gap validation.

We do not reuse earlier baseline/stress/hard labels. Across all 36 prespecified targets, higher repetition generally increases attack AUC, particularly for Z and ZZ encoders; increasing depth also raises leakage relatively. Gap and loss-AUC have Pearson $r = 0.948$ (CI [0.864, 0.982]) and Spearman $\rho = 0.931$ (CI [0.710, 0.974]). This removes attack-outcome selection within the factorial and supports gap as a strong descriptive proxy, not a deterministic predictor.

3. Factor attribution.

Fig. 8 is a dataset-wise normalized factor-association analysis of the generalization-gap proxy. As detailed in the full response link, categorical terms use a bounded one-way ANOVA-derived score, while numeric and product terms use absolute Pearson correlation. These scores are then normalized to sum to 100. Thus, a reported percentage represents a share of the aggregate displayed association score. The new paired MIA interval results separately corroborate the feature-map/repetition associations.

4. Architecture controls.

We evaluated QNN/HQNN/QCNN and a classical MLP with a comparable small parameter budget, reporting exact parameter/gate counts. Relative to QNN, MLP test accuracy changes by +0.146 $\pm$ 0.093 (CI [0.062, 0.259]) with unresolved gap/AUC effects. QCNN changes accuracy by +0.190 $\pm$ 0.052 (CI [0.148, 0.249]), gap by −0.041 $\pm$ 0.042 (CI [−0.083, −0.007]), and loss-AUC by −0.019 $\pm$ 0.025 (CI [−0.040, −0.001]). This supports architecture as a moderator, though wrappers retain different components and are not matched causal ablations.

5. Baselines and scope.

We expanded attack evaluations spanning access models. Online LiRA with fixed variance proved strongest (AUC 0.609 $\pm$ 0.063, TPR@10% FPR 0.192 $\pm$ 0.069), while loss-threshold (AUC 0.596 $\pm$ 0.052, TPR@10% FPR 0.135 $\pm$ 0.039) and label-only attacks preserve structural ordering.

We expanded attack evaluations across access models, all of which recovered the same aggregate feature map and repetition trends. Fixed-variance online LiRA achieved the highest mean paired MIA interval (AUC 0.609 $\pm$ 0.063, TPR@10% FPR 0.192 $\pm$ 0.069), followed by loss-threshold (AUC 0.596 $\pm$ 0.052, TPR@10% FPR 0.135 $\pm$ 0.039), while label-only confirmed these trends using predicted labels alone.

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Throughout, we consider a standard black-box setting where adversaries observe only classical outputs. However, whether a quantum-capable adversary could leverage direct quantum measurements (e.g., finite-shot statistics) as inputs for an attack remains a future direction for us. Finally, references to domains such as healthcare, finance, and genomics are intended to motivate why privacy deserves attention as QML matures; hence, we narrow our claims in the revision.

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